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Impact of wearable device-based interventions with feedback for increasing daily walking activity and physical capacities in cardiovascular patients: a systematic review and meta-analysis of randomised controlled trials

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ABSTRACT

Objective To systematically review randomised controlled trials (RCTs) using a wearable physical activity monitoring device as an intervention to increase daily walking activity and improve physical capacities in patients with cardiovascular disease (CVD).

Design Systematic review and meta-analysis of RCTs.

Data sources PubMed, Embase and Web of Science from inception to June 2022.

Eligibility criteria Randomised controlled studies including patients with CVD over 18 years of age at the end of a cardiac rehabilitation programme comparing an intervention group using a wearable physical activity monitoring device with feedback with usual care or with a control group receiving no feedback on their physical activity and reporting a change in the daily number of steps and/or a change in the distance covered in the 6-minute walk test (6-MWT) or a change in peak oxygen uptake ($VO2_{peak}$) as endpoints.

Results Sixteen RCTs were included. The intervention of wearing a physical activity monitoring device with feedback significantly improved daily number of steps compared with controls (standardised mean difference (SMD) 0.85; 95% CI (0.42; 1.27); p<0.01). The effect was greater when the duration of the intervention was less than 3 months (SMD 1.0; 95% CI (0.18; 1.82); p<0.01) than when the duration of the intervention was 3 months or longer (SMD 0.71; 95% CI (0.27; 1.16); p<0.01), but no significant interaction was found between subgroups (p=0.55). 6-MWT distance and $VO2_{peak}$ showed only small effects (SMD 0.34; 95% CI (−0.11; 0.80); p=0.02 and SMD 0.54; 95% CI (0.03; 1.03); p=0.07, respectively).

Conclusion The use of wearable physical activity monitoring devices appears to help patients with CVD to increase their daily walking activity and thus their physical activity, particularly in the short term.

PROSPERO registration number CRD42022300423.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This is a systematic review and meta-analysis of randomised controlled trials using a wearable physical activity monitoring device as an intervention to increase daily walking activity and improve physical capacities in patients during or after cardiac rehabilitation.
- ⇒ A protocol was registered on PROSPERO and methods were reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement.
- ⇒ Methodological quality of the studies was assessed using the ROB2 tool.
- ⇒ The standardised mean difference was used to calculate the effect sizes of each outcome to consider the different types of wearable activity monitors used and the different lengths of intervention.

INTRODUCTION

Cardiovascular diseases (CVDs) remain the leading cause of mortality and disability accounting for approximately 32% of all deaths worldwide[1] and have a significant impact on healthcare costs and systems. Since 2011, physical activity has been recognised as an entire part of therapeutic management in primary, secondary and tertiary prevention; and the benefits of physical activity on morbidity/mortality and health no longer need to be demonstrated.[1–3] The initiation and practice of regular physical activity of moderate intensity would reduce the risk of cardiovascular mortality by 30%[1 3] and would contribute to a significant improvement in quality of life, an increase in life expectancy, a reduction of the number of hospital admissions and a reduction in loss of autonomy.[1 4] Despite the important proven benefits of physical activity and guidelines recommendations, only 30% of patients treated for a cardiovascular event (myocardial infarction, coronary

revascularisation or heart failure) take part in a physical rehabilitation programme for 3 months in a cardiac rehabilitation centre.[5–8] Several factors may explain the reluctance to engage in physical activity, such as inconvenient programme schedules, transportation restrictions, poor accessibility to practice sites and infrastructures, physical limitations or psychological barriers,[9–11] as well as lack of motivation and time.[12] Strengthening patients' adherence to physical activity appears to be an essential issue for personalised management of CVD, an issue aimed at improving the patient's quality of life by considering his/her lifestyle in its entirety.[13 14] Wearable physical activity monitoring devices (ie, smartwatch, pedometer or fitness band) provide real-time feedback of various measures regarding physical activity, such as tracking the number of daily steps, activity time and distance walked. These devices have the advantage of improving adherence to physical activity by allowing users to receive feedback on their physical activity and monitor their own progress.[15] Several studies have shown that the use of wearable devices recording physical activity is an effective technology to encourage[16] and increase physical activity, whether in the general population,[17] after stroke,[18] in patients with coronary heart disease or heart failure,[19] or in patients with cardiac pathology.[20] Along with behaviour change techniques (BCTs), wearable physical activity monitoring devices may be effective to provide an increase in physical activity. In two meta-analyses,[21 22] the authors tried to determine which elements were most likely to increase motivation and physical activity. They found that techniques such as planning, goal setting, self-monitoring, feedback, positive reinforcement and objective benefits perception were key components of an effective behaviour change intervention. These components were also found in a meta-analysis examining the effectiveness of 30 interventions to increase physical activity including goal setting, self-monitoring and social support was associated with better effectiveness in promoting changes in diet and/or physical activity (eg, frequency, metabolic equivalent-hours per week).[23] Wearable physical activity monitoring devices along with feedback and a daily steps goal may be an easy and effective way to increase daily physical activity in patients with CVD as they are accessible to everyone, do not require specific skills or infrastructures, and can be easily added into everyday life.

Walking can be implemented in daily life without difficulty[24] and seems to be one of the best opportunities to increase patient adherence to regular advice to maintain physical activity. Studies have shown that walking increases heart rate, improves blood flow and reduces blood pressure; and a meta-analysis also showed that walking increased aerobic capacity, lowered blood pressure, and reduced body mass index and body fat.[25] This was supported by a randomised controlled trial (RCT) in adults with uncontrolled hypertension placed on a diet and walking regimen. Intervention group adults reduced their systolic blood pressure by increasing their steps per day by 2000.[26]

Combining walking activity and wearable physical activity monitoring devices appears to be the future alternative to centre-based cardiac rehabilitation, as physician shortages become a reality and the location of cardiac rehabilitation facilities does not always match the patient's geographical location.

Previous systematic reviews and meta-analyses of general population patients with chronic diseases have shown that interventions including wearable physical activity monitoring devices increased the amount of daily physical activity[27–30] but did not focus only on patient population with CVD during or after cardiac rehabilitation nor took into account the length of the intervention.

Therefore, we conducted a systematic review and meta-analysis of RCTs that tested the use of wearable physical activity monitoring devices with feedback in patients during or after cardiac rehabilitation to assess whether this intervention increases daily walking activity in the short and medium terms.

METHODS

The meta-analysis protocol was prospectively registered in the International prospective register of systematic reviews (PROSPERO, https://www.crd.york.ac.uk/PROSPERO, registration number: CRD42022300423). This meta-analysis was written according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses statement for reporting.[31]

Patients and public involvement

It was not possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Search strategy

The systematic search included the following databases: PubMed, Embase and Web of Science. These databases were searched from inception to June 2022 with no restriction on language or publication period. The search strategy involved four types of terms in English: "cardiovascular disease", "activity tracker", "activity monitoring", "physical activity" combined with the Boolean "AND" and "OR" under different combinations and different synonym keywords and Medical Subject Headings terms. The search terms were done in all fields. The full search strategy is available in online supplemental table 1.

Study selection

Two researchers (A-NH and CLH) independently performed the systematic search in three different databases (PubMed, Embase and Web of Science) and independently assessed eligibility of the retrieved records. The study selection process was performed based on two successive steps: (1) by reading the titles and abstracts, and (2) by reading the full texts. The search results were entered into a reference management tool (Zotero V.6.0.8) and duplicates from different databases were removed. The

two reviewers analysed the inclusion and non-inclusion criteria for each study, and discrepancies were discussed until a consensus was reached. To be eligible, studies had to consist of the following: (1) to be RCTs; (2) to include patients of any sex older than 18 years with CVD; (3) to compare the intervention using a wearable physical activity monitoring device with usual care control group receiving no feedback on their physical activity (usual care defined as routine care received by patients in secondary and tertiary prevention and management of CVD; this typically includes physical activity advice, multicomponent disease management advice (physical activity practice, nutrition and diet, etc) and is often associated with cardiac rehabilitation programmes); and (4) to report measures assessing physical activity such as change in number of steps per day and/or change in walking distance covered in the 6-minute walk test (6-MWT) or change in peak oxygen uptake ($VO2_{peak}$).

The goal of the intervention was to increase or maintain physical activity levels. The wearable physical activity monitors were to be equipped with an accelerometer and/or pedometer. Intervention times and duration of cardiac rehabilitation programmes were taken into account.

Abstracts, poster presentations, conference presentations, unpublished books and letters to the editor or book chapters not reporting RCTs were not included.

Studies that did not use a wearable monitoring device as an intervention to increase physical activity were not included. Studies in which participants in the intervention group did not receive feedback on their physical activity data, as well as studies in which control groups had access to or received feedback on their physical activity data, were not included.

Data extraction and quality assessment

Data were extracted by A-NH following a data extraction template designed for this meta-analysis and verified by a second author (CLH). For each study, the following data were extracted: (1) details on study characteristics (authors' names, year of publication, journal, title, country where the study was conducted, number of patients randomised in each group); (2) study design; (3) methodological quality of the study; (4) patient's characteristics (mean age, sex, pathology); (5) intervention characteristics (presence or absence of a control group, duration, objective, type of activity tracker); and (6) outcome measures. In order to compute the mean change and mean SD according to the method described in Chapter 8 of the Cochrane Handbook for Systematic Reviews of Interventions,[32] the following data were extracted: number of randomised patients in each group, mean value pre-intervention and post-intervention in each group, SE and CI.

The methodological quality of each trial was evaluated using the Cochrane Collaboration tool for assessing the risk of bias in RCTs according to five domains of bias: bias arising from the randomisation process; bias due to deviations from intended interventions; bias due to missing outcome data; bias in measurement of the outcome; and bias in selection of the reported result.[33] The risk of bias was assessed by two independent authors (A-NH and CLH) according to the ROB tool on three different outcomes (steps count, 6-MWT distance, $VO2_{peak}$). If disagreements came out, a consensus was reached through discussion.

Outcomes

The primary outcome was change in number of steps per day. Secondary outcomes were change in distance walked in the 6-MWT and $VO2_{peak}$. All outcomes were calculated as the average difference of the outcome (number of steps per day, distance walked in the 6-MWT and value of the $VO2_{peak}$) before and after the intervention in each group.

Statistical analysis and synthesis

Effect sizes were calculated as the difference in outcomes between the baseline and end-of-intervention values using the standardised mean difference (SMD). SMD was used in order to standardise the results of the studies because although the studies all assessed the same outcomes, they measured them in different ways due to the different types of wearable activity monitors used and the different lengths of intervention. SMDs were calculated by dividing the mean differences by their respective SDs allowing to express the size of the intervention effect in each study relative to the variability observed in that study. The interpretation of SMD values was as follows: large effect when the SMD was ≥0.8; moderate effect when the SMD was between 0.5 and 0.79; and small effect when the SMD was between 0.2 and 0.49. When the mean change and SD of change were missing, we followed the calculation method described in Chapter 8 of the Cochrane Handbook for Systematic Reviews of Interventions[32] using a correlation coefficient derived from a study providing the mean change and SD of change as recommended.[34]

Heterogeneity was assessed using I^2 and was considered significant if $p>0.1$. Heterogeneity was considered as minimal and might not be important if $I^2<40\%$, moderate if I^2 was between 30% and 60%, substantial if I^2 was between 50% and 90% and considerable if I^2 was >90%.[35] Random-effects model with SMD was used due to the variability of the assessment and duration of the intervention across the included studies.[35]

The results of the meta-analysis are presented graphically with forest plots representing the change in number of steps per day, $VO2_{peak}$ and 6-MWT expressed as the mean differences with corresponding 95% CI between intervention and control groups. One study[34] included two different intervention groups using a wearable activity monitor and two different control groups. We combined the two intervention groups and the two control groups following the recommendations and guidelines in the Cochrane Handbook for Systematic Reviews of Interventions (Chapter 6, Section 6.5.2.10).[32]

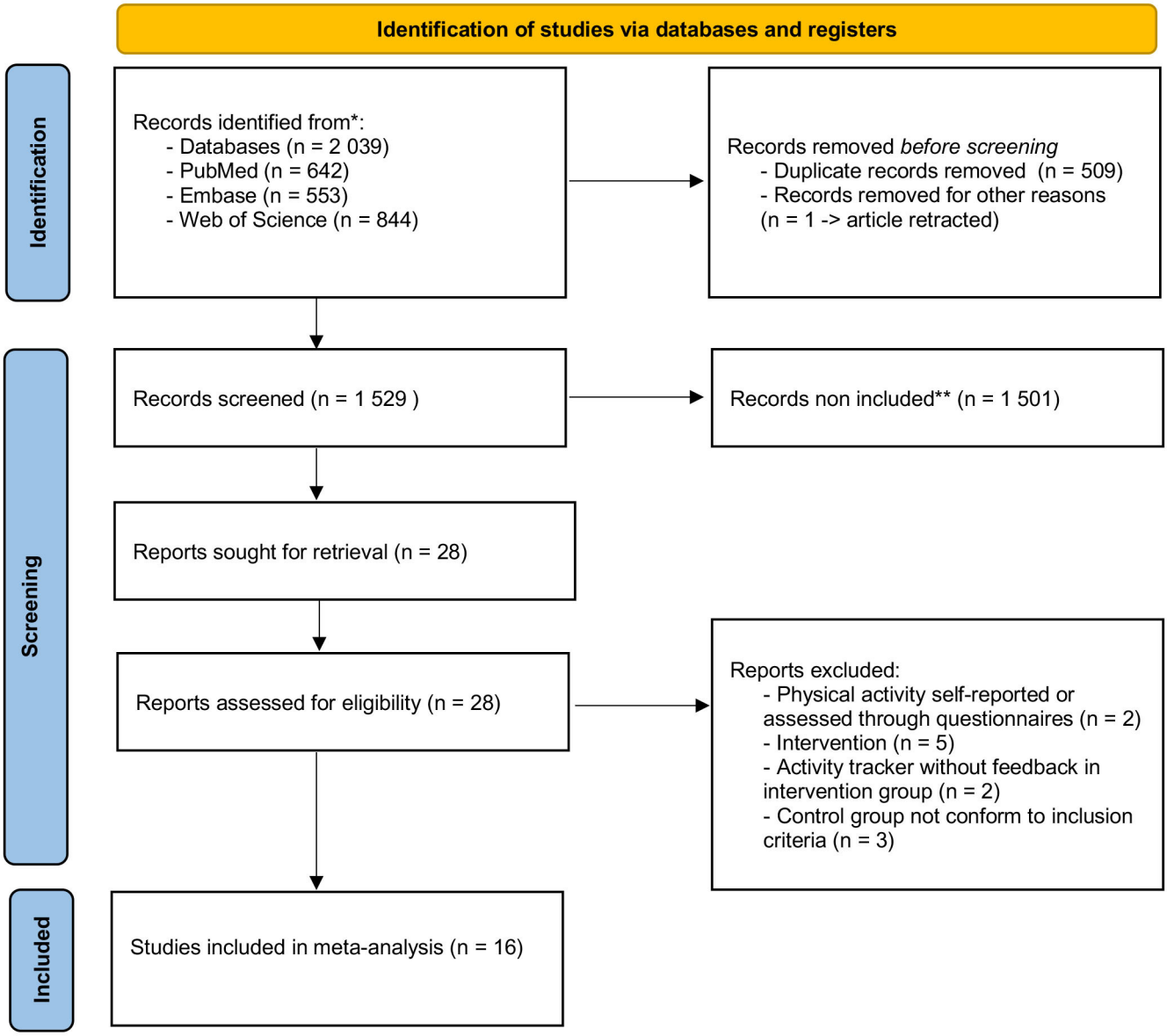

Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analysis diagram for the search and selection process.

Funnel plots were created for any outcomes to explore the publication biases of the studies included in the meta-analysis.[36]

Consistency of results was verified through subgroup analysis according to the duration of the interventions: <3 months and ≥3 months. The meta-analysis was performed using R Studio (V.1.4.1103) and dmetar package.[37]

RESULTS

Study selection

The search procedure identified 2039 articles of which 1529 were reviewed after removing duplicates. After review of titles and abstracts, 1501 additional articles were not included. After evaluation of the full texts, 12 additional references were not included. Finally, 16 articles were included in the meta-analysis, yielding a total of 1427 patients (figure 1).

Characteristics of the patients

The patient characteristics of the included studies are summarised in table 1. The included studies were published between 2011 and 2022.

The total number of participants across studies was 1427 and ranged from 18[38] to 347.[39] There were 732 patients in the intervention groups and 695 patients in the control groups. The percentage of men in all trials was 67% (957 male participants). The mean age of all participants was 62.5±9.9 years and ranged from 53.3 years[38] to 76 years.[40] The mean age in the intervention groups was 61.6±9.8 years ranging from 53.3 years[38] to 76 years.[40] The mean age in the control groups was 63.4±10.1 years ranging from 58.2 years[41] to 76 years.[40]

Intervention groups

The intervention consisted of wearing a wearable physical activity monitor and receiving daily step goals, a

Table 1 Characteristics of studies

Study	Year	Country	Design	Number of participants	Mean age intervention	Mean age control	% male intervention	% male control	% dropout	CVD type
Houle *et al*[47]	2011	Canada	RCT	65	58.0	59.0	81.3	75.8	20.7	Coronary artery disease
Izawa *et al*[46]	2012	Japan	RCT	126	59.2	59.1	78.8	82.3	18.2	Heart disease, coronary artery bypass grafting, myocardial infarction, valve replacement
Kaminsky *et al*[38]	2013	USA	RCT	18	53.3	59.4	80.0	75.0	28.6	Coronary artery disease, myocardial infarction, heart surgery
Martin *et al*[45]	2015	USA	RCT	48	56.5	60.0	53.1	56.3	?	Hypertension
Mansfield *et al*[42]	2015	Canada	RCT	60	64.0	61.5	69.0	57.0	5.0	Stroke without walking limitations
Frederix *et al*[48]	2015 (ePub 2013)	Belgium	RCT	80	58.0	63.0	81.0	85.0	17.5	Coronary artery disease
Frederix *et al*[49]	2015	Belgium	RCT	139	61.0	61.0	85.5	78.6	10.0	Coronary artery disease, chronic heart failure
Danks *et al*[41]	2016	USA	RCT	37	59.1	58.2	36.8	44.4	5.4	Stroke without walking limitations
Dasgupta *et al*[39]	2017	Canada	RCT	347	60.0	59.4	43.1	91.0	21.0	Hypertension
Duscha *et al*[51]	2018	USA	RCT	32	59.9	66.5	81.2	66.7	21.9	Coronary artery disease
McDermott *et al*[50]	2018	USA	RCT	200	70.1	70.4	45.5	49.5	9.0	Peripheral artery disease
Kanaï *et al*[43]	2018	Japan	RCT	57	66.8	62.9	65.2	52.0	12.7	Stroke without walking limitations
Ozemek *et al*[34]	2020	USA	RCT	99	61.7	61.2	73.6	82.6	No information	Coronary artery disease, myocardial infarction, heart surgery
Grau-Pellicer *et al*[44]	2020	Spain	RCT	41	63.0	54.2	67.0	47.1	17.1	Stroke without walking limitations
Lindman *et al*[40]	2021	USA	RCT	50	76.0	76.0	64.0	68.0	0.0	Aortic valve stenosis, transcatheter aortic valve replacement
Nagatomi *et al*[52]	2022	Japan	RCT	30	59.8	67.7	60.0	47.0	0.0	Chronic heart failure with physical frailty

CVD, cardiovascular disease; RCT, randomised controlled trial.

physical activity prescription or exercise training for the duration of the intervention. The duration of the intervention varied between studies: the duration was <3 months in seven studies[38 40 42–46] and ≥3 months in nine studies[34 39 41 47–52] (online supplemental table 2). In the intervention groups, general advice and guidelines were supplemented with self-monitoring feedback and/or goal setting, and/or encouragement to increase daily physical activity (number of steps, amount of physical activity, duration of walking sessions), and/or personalisation of the rehabilitation programme to improve self-confidence and self-regulation.

Control groups

The advice given to patients in the control groups was to exercise according to common guidelines (30–40 min of moderate to vigorous physical activity per day or just stay active). No one received feedback on their activity and no one used a wearable physical activity monitoring device to self-monitor their activity.

Risks of bias

The risk of bias in the studies is presented in the online supplemental figures 1 and 2. The number of unclear verdicts was high because the supposedly blinding status of participants or assessors was not always complied with. Two studies were considered to have a high risk of bias: the study by Grau-Pellicer *et al*[44] which raised some high-risk flags as nearly half of the participants in the intervention group did not use the app for multiple reasons and only participants with a high level of adherence in the intervention group were compared with the control group, suggesting a failure in the implementation of the intervention; and a high level of risk of bias was also found in Kaminsky *et al*'s study because of a lack of details on the randomisation process.[38]

Effects of the intervention

Primary outcome: change in number of steps per day

The number of steps before and after intervention was calculated in 11 of the included studies (884 patients)[34 38–43 45–47 51] (figure 2). The use of a wearable physical activity monitoring device significantly increased the number of daily steps compared with the control groups (SMD 0.85; 95% CI (0.42; 1.27); p<0.01; I^2=86%; mean difference (MD) 1534; 95% CI (843; 2225); p<0,01; I^2=85%). The effect of using a wearable physical activity monitoring device on daily steps was greater when the duration of the intervention was <3 months (SMD 1.0; 95% CI (0.18; 1.82); p<0.01; MD 1636; 95% CI (474; 2797); p<0.01) compared with ≥3 months (SMD 0.71; 95% CI (0.27; 1.16); p<0.01; MD 1410; 95% CI (600; 2220); p<0.01); but no significant interaction was found between the subgroups (p=0.55). The funnel plot was symmetrical, arguing for a lack of publication bias (online supplemental figure 3).

Secondary outcomes: distance walked during the 6-MWT and $VO2_{peak}$

Six-minute walk test

The walking distance covered in the 6-MWT was reported in five studies[40 41 44 50 52] (253 patients) (figure 3). The positive effect of wearable device intervention on improvement in 6-MWT performance was small (SMD 0.34; 95% CI (−0.11; 0.80); I^2=65%; p=0.02). The funnel plot was symmetrical, arguing for a lack of publication bias (online supplemental figure 4).

A sensitivity analysis excluding the study by McDermott *et al*,[50] which included patients with peripheral arterial disease with an initial walking distance <500 m, showed a moderate positive effect of wearable devices on 6-MWT performance (SMD 0.51; 95% CI (0.03; 0.99); I^2=50%; p=0.11) (online supplemental figure 5).

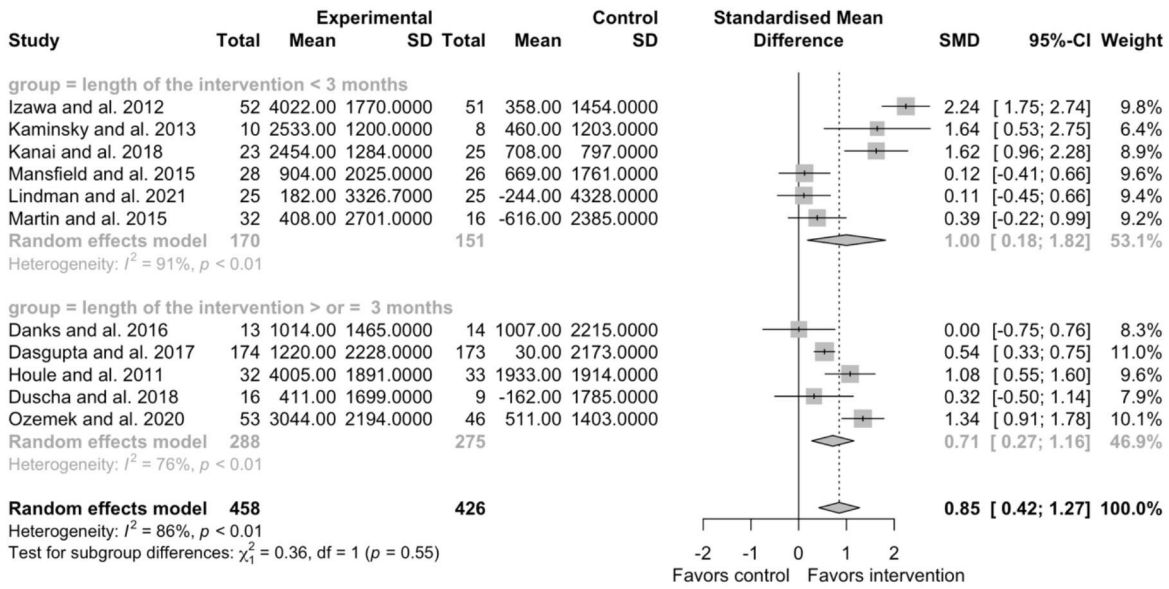

Figure 2 Forest plot of daily steps outcome. SD, standardised difference; SMD, standardised mean difference.

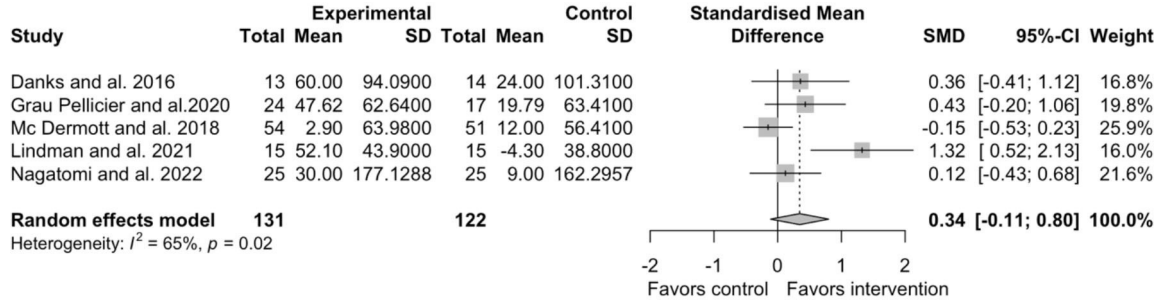

Figure 3 Forest plot of 6-minute walk test outcome. SD, standardised difference; SMD, standardised mean difference.

Change in $VO2_{peak}$

Three studies reported $VO2_{peak}$ results[48 49 51] (226 patients, figure 4). The forest plots showed a moderate effect with the use of wearable monitoring devices (SMD 0.54; 95% CI (0.03; 1.03); I^2=65%; p=0.06) (figure 4).

DISCUSSION

The objective of this meta-analysis was to quantify the effectiveness of wearable physical activity devices in increasing daily walking activity and improving physical abilities in patients with CVD (coronary artery disease, valve replacement, stroke, peripheral arterial disease, hypertension). Results suggest that the use of wearable devices helped such patients increase their daily step counts (SMD 0.85; 95% CI (0.42; 1.27); p<0.01; I^2=86%) and therefore their daily physical activity. This suggests that the use of wearable physical activity monitoring devices with feedback may help cardiovascular patients to increase their daily walking, overcoming common physical activity practice barriers such as inappropriate scheduling of proposed physical activity, lack of time and lack of infrastructures to practise. These results are consistent with a previous systematic review and meta-analysis: in patients with chronic CVD using a wearable device measuring physical activity, the number of daily steps significantly increased by 1300 steps/day[27] and by 1656 in patients with cardiometabolic diseases (obesity, diabetes mellitus and CVD).[30]

A recent meta-analysis of three RCTs of supposedly healthy subjects and patients (any human population of any age) using a pedometer or accelerometer also showed a significant increase in daily steps (n=211, +2587 steps/day (95% CI: 916 to 5257); I^2=74.6% and p=0.002).[53] An umbrella review showed an increase of 1800 steps/day and an increase of 40 min of walking per day in clinical and non-clinical populations.[29] Along with those results and the average increase in steps per day, it seems that wearable physical activity monitoring device-based intervention has a positive impact on daily walking. This supports our meta-analysis result as we found a mean increase of 1534 steps/day in the intervention group.

A systematic review including more than 30 000 non-diseased adults reported that an increase of 1000 steps/day reduced all-cause mortality by 6%–36% and reduced CVD morbidity and mortality by 5%–21%.[28] This suggests that the use of wearable physical activity monitoring devices to help increase daily walking activity may help indirectly reduce CVD morbidity and mortality.

Wearable physical activity monitoring devices propose a wide range of BCTs that are typically used in clinical behavioural intervention such as goal setting, self-monitoring and feedback.[23 54 55] As implementing an intervention with BCTs without wearable physical activity monitoring device requires BCTs formed staff and an individual patient follow-up, wearable physical activity monitoring device seems to be a device by which intervention using BCTs may be translated into widespread use and facilitate the implementation of such programmes.[54] In the present meta-analysis, interventions were not solely based on the use of a wearable physical activity device and sometimes also included other BCTs. The use of BCTs such as coaching or encouragement is simplified with the use of wearable physical activity monitoring devices as they provide concrete data on physical activity. These data can be used to set goals in coaching, or to encourage the users through feedback on their activity and on what they have already achieved. In addition, several BCTs are included in wearable physical activity monitoring device and provide guidance and support for physical activity through self-monitoring, goal setting

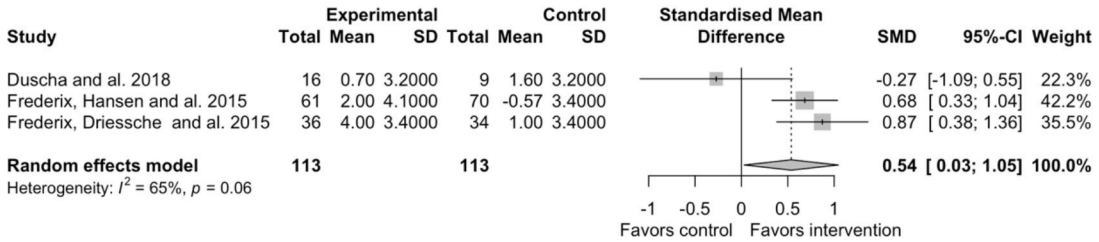

Figure 4 Forest plot of peak oxygen uptake outcome. SD, standardised difference; SMD, standardised mean difference.

and feedback.[55 56] Self-monitoring allows for more accurate coaching as physical activity metrics are objective and allows for personalised and adapted goal setting. Feedback on physical activity throughout the day allows for appropriate coaching and encouragement based on current physical activity. Self-monitoring through wearable devices appears to be a crucial component of BCTs that promote an increase in physical activity,[57 58] but it is still unclear whether using wearable devices alone is sufficient to elicit long-term behaviour change or if it depends on a particular combination of BCTs with wearable devices.[15 59] However, there are limited available data on the additional benefits of other BCTs to self-monitoring through wearable devices. According to a recent meta-analysis, multicomponent interventions that combine self-monitoring through wearable devices with other BCTs provide additional benefits beyond self-monitoring.[60] Therefore, it is possible that other components, such as encouragement and personalised coaching, may explain some part of the heterogeneity observed along with the interventions through wearable devices.

The positive impact of wearable physical activity monitoring devices may be stronger when the intervention length is <3 months[61] due to the novelty effect: the use of a physical activity monitoring device changes patients' routines; at the beginning they adhere strongly to it and seek feedback more often, then after the novelty effect has worn off, this effect may also depend on the initial patient's motivation.[61] It is possible that after a certain time of intervention, patients will no longer increase their physical activity[62]: they have reached a plateau in increasing their physical activity and so both self-monitoring and coaching no longer help them to increase it. Yet, this does not mean that the combination of coaching, encouragement and self-monitoring could not work to help maintain and/or persist in physical activity. A review of fitness technology for increasing physical activity stated that interventions supported with additional motivational techniques appeared to be more effective in facilitating long-term changes in behaviour.[15] Nevertheless, the literature on this subject is limited, and it is difficult to determine which combination of BCTs and self-monitoring is essential to long-lasting change in physical activity.

A moderate effect was found on $VO2_{peak}$, but only three studies in the meta-analysis provided this measure. This result is in line with Hannan *et al*[63] who showed that wearable activity monitoring devices with exercise prescription or advice significantly improved $VO2_{peak}$ compared with interventions not using wearable devices. This is an interesting finding for the impact of wearable devices on cardiorespiratory fitness, as one study showed that a 1 mL/kg/min higher $VO2_{peak}$ was associated with a 9% reduction in the relative risk of all-cause mortality.[64] Yet, the SMD CI ranged from no effect (0.03) to large effect (1.03), making it difficult to draw a conclusion and affects our certainty of the evidence. Further information would be needed to conclude on wearable device impact on cardiorespiratory fitness. This wide variation may be explained by the limited number of studies included in the meta-analysis of maximal oxygen consumption, intervention variety and variable adherence to interventions. The positive effect on $VO2_{peak}$ is likely due to an increase in adherence to exercise prescription rather than an increase in walking time. Our meta-analysis has shown a small to moderate effect on distance walked during 6-MWT. However, we must be careful about the interpretation as the SMD CI ranged from no effect to large effect. A second analysis removing McDermott *et al*[50] as patients had a walking distance <500 m at the beginning of the study did not drastically change the results for this outcome and the effect remained moderate. Results of Varnfield *et al* showed that when comparing results of a rehabilitation programme performed at home with a smartphone with the results of a rehabilitation programme performed at a centre in patients with prior event of myocardial infarction, covered distance during 6-MWT was not significantly different between the two groups.[65] This may be explained by the various lengths of the interventions, as interventions lasting longer may have helped patients to walk more. It may also be due to the fact that the 6-MWT is not the best way to assess efficiency of wearable devices, but it could also suggest that it depends on the motivation/adherence of the participants to the programme: the more they adhere, the more they will be able to increase their physical capacities and walking abilities. This could be supported by Evangelista *et al*[66] who found that patients adhering more to the programme over 6 months also improved their 6-MWT. All this helps to explain the large range from no effect to large effect of the SMD CI.

Demographics

Demographics showed that 67% of the participants in the trials were male patients, suggesting to be careful and not to generalise to all patients with CVD and especially to female patients as they may experience CVD in a different way compared with male patients.[67 68] Indeed, some studies showed significant differences between males and females: a lower rate of entering cardiac rehabilitation programme in female patients and a higher percentage of female patients dropped out.[69 70] These studies confirmed the need to address sex differences in physical activity research and to be careful about generalisation of the present meta-analysis results. The majority of patients included in the studies were Caucasian, with the majority of studies taking place in North America (n=10), Europe (n=3) or Japan (n=3). In the USA, healthcare insurance is not affordable for everyone because the costs are very high, leading to inequality in health. These inequalities in healthcare also reflect the general ethnic inequity of the country and explain the high rate of Caucasian patients included in the studies. Therefore, the generalisation of this meta-analysis is limited and further research should highly consider enrolment of all ethnic types.

Based on these elements and on the results of this meta-analysis, we could only conclude that wearable physical activity monitoring devices with feedback seem to be effective as part of intervention to increase daily walking activity. Future studies should focus and explore which combination of BCTs and wearable physical activity monitoring device intervention is the most effective. Some previous research began to answer this question. In their systematic review, Schoeppe *et al*[71] provided modest evidence that app-based interventions could improve physical activity and that multicomponent intervention appeared to be more effective than intervention using only 'standalone' app. This systematic review, along with the results of our meta-analysis, suggests continuing research to identify the key component to add in wearable physical activity monitoring devices to maximise physical activity increase. It is important to understand the mechanism underlying the effect of the efficacy of wearable physical activity devices as it could help improve future interventions and global efficacy, not only on physical activity but also on overall health including well-being, self-esteem and self-efficacy.

Wearable physical activity monitoring devices are likely to help patients with CVD to increase their daily activity even if they do not attempt cardiac rehabilitation programmes. They may be great tools to personalise interventions and make the patient truly an actor of his/her own health, increasing motivation and adherence to physical activity programmes.[72]

As stated above, it will be interesting to investigate the factors influencing adherence, effectiveness and mechanisms underlying the increase in physical activity. With this, it would be possible to design specific remote intervention to be more effective in increasing and maintaining physical activity in order to improve patients' global health. Use of wearable physical activity monitoring devices is part of the digital health 'revolution'. The WHO states that 'eHealth has played a key role in improving the quality of services, increasing coordination between providers, improving patient management, helping to overcome physical distances between patients and providers and engaging patients in their own health and well-being'.[73] Remote/digital healthcare delivery will be needed more and more as demographics, population health and prevalence of chronic disease will continue to exert pressure on healthcare resources. As wearable physical activity monitoring devices enable remote rehabilitation and are now mature and well documented, they should be opening a new part of healthcare delivery solution engaging the patients and placing them in the centre of their own health.[72]

Limits of the present meta-analysis

Moderate to high statistical heterogeneity was detected across the studies. This may be explained by the variety of wearable monitoring devices used, and the different numbers and combinations of intervention and BCTs in the intervention groups: some used smartphone applications in combination with a wearable physical activity monitoring device, text messaging or counselling sessions making it difficult to draw a single kind of intervention and to investigate the isolated effect of wearable physical activity monitoring devices on physical activity.

Also, the feedback was present in each intervention group as specified in the inclusion criteria and is recognised as an important component in BCTs.[74] Most of the included studies also proposed some goal setting, which is an additional important intervention component in BCTs.[75] Combining these two components and personalising it to the patient seem to be the most appropriate way to elicit some behaviour changes and may be a lot more effective to promote physical activity compared with general physical activity advice. It would have been interesting to perform subgroup analyses according to the wearable devices used and the number of BCTs included in the interventions. However, due to the diversity of wearable devices used in the studies and the BCTs employed, we lacked statistical power to perform such subgroup analyses.

In addition, the frequency of feedback on physical activity metrics provided to patients and/or study investigators differed between studies. In some studies, patients had continuous access to feedback on their physical activity through a smartphone application or a fitness tracker with display. In others using pedometers or accelerometers, patients were asked to log their activity daily allowing them to have at least one feedback per day and even up to three feedback per day in one study. The way in which physical activity metrics were accessible and the easy-to-read feedback may have had a significant impact on the effectiveness of wearable physical activity monitoring device intervention. The frequency of feedback and how it is provided should be carefully considered in future studies as it is a key element in the practice and maintenance of physical activity.

Differences in the control groups included may explain some of the heterogeneity observed since some of the studies included a usual care control group, while others provided general physical activity advice or face-to-face sessions on physical activity lifestyle factors or motivational messages to the control group. However, a majority of the included studies set a usual care control group and none had feedback on activity nor used a wearable physical activity monitoring device to self-monitor activity. Another point about control groups is that some control group patients had a blinded wearable physical activity monitoring device. This may have influenced the physical activity patterns and habits of these patients as they knew their physical activity was recorded. This phenomenon is referred to as reactivity or the Hawthorne effect. Studies on accelerometer reactivity have shown that it was not present across the evaluated populations (children, teens, healthy adults or patients with heart failure).[76 77] Furthermore, as patients were encouraged to practise physical activity even when they do not wear a wearable device and know that the purpose of the study is to observe physical activity, they will probably also increase their activity at

the beginning of the intervention. This is similar to the placebo effect observed in studies evaluating medical treatment. Yet, if changes were present, they probably would not last through the intervention length as habits are difficult to counter and they did not receive any feedback on their physical activity.

Finally, if simply knowing that physical activity metrics will be recorded increases physical activity, this goes in the way of the potential 'efficacy' of wearable devices as just wearing them and knowing physical activity metrics are recorded would increase physical activity.

Conclusion

Interventions using wearable physical activity monitoring devices with feedback to help patients with CVD increase their daily walking activity may be effective at the end of cardiac rehabilitation at short (<3 months) and medium (≥3 months) terms when referring to mean daily steps.

Our meta-analysis focused on the impact of wearable physical activity monitors used directly post-rehabilitation and not on their impact if used long after the end of a rehabilitation programme. Therefore, we could not know if the increased physical activity lasts after the end of the wearable physical activity monitoring device intervention and, if not, how long it is maintained. This suggests that more studies are necessary to investigate the long-term sustainability of these positive effects and to be careful on the conclusions we draw especially as the heterogeneity was high and the effects small to moderate.

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